

Member vs. Casual Riders: A Cyclistic Bike-Share Analysis

Understanding behavioural differences to guide a casual-to-member conversion strategy

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3.87M

rides analysed

3×

longer casual rides

77%

of rides by members

The bottom line

Members are commuters; casual riders are leisure users. Members take short trips (12.9 min avg) that peak midweek. Casual riders take trips ~3× longer (39.6 min avg) that peak on weekends and surge in summer, starting from lakefront and tourist stations. A conversion campaign should therefore sell weekend and recreational value — not commuter savings — timed to spring/summer and marketed where casual demand concentrates.

The question

Cyclistic's marketing team wants to convert casual riders into annual members, who are more profitable. To design that campaign they first need to know: **how do annual members and casual riders use the bikes differently?**

Approach

Source. A full year of public Divvy trip data — four quarterly files (Apr 2019 – Mar 2020), 3.87M rides. The files arrived in three different schemas (clean column names, verbose labelled headers, and a redesigned 2020 layout), so the core prep task was conforming all three: mapping usertype (Subscriber/Customer) to member_casual, aligning column names, and recalculating ride length from timestamps since the 2020 file has no duration field.

Tools. Built entirely in Power BI — Power Query (M) to clean and append the files, DAX measures to compare the groups, and Power BI visuals to tell the story.

Derived fields. Added the two calculated columns the analysis depends on: ride_length (ended_at minus started_at) and day_of_week (from started_at).

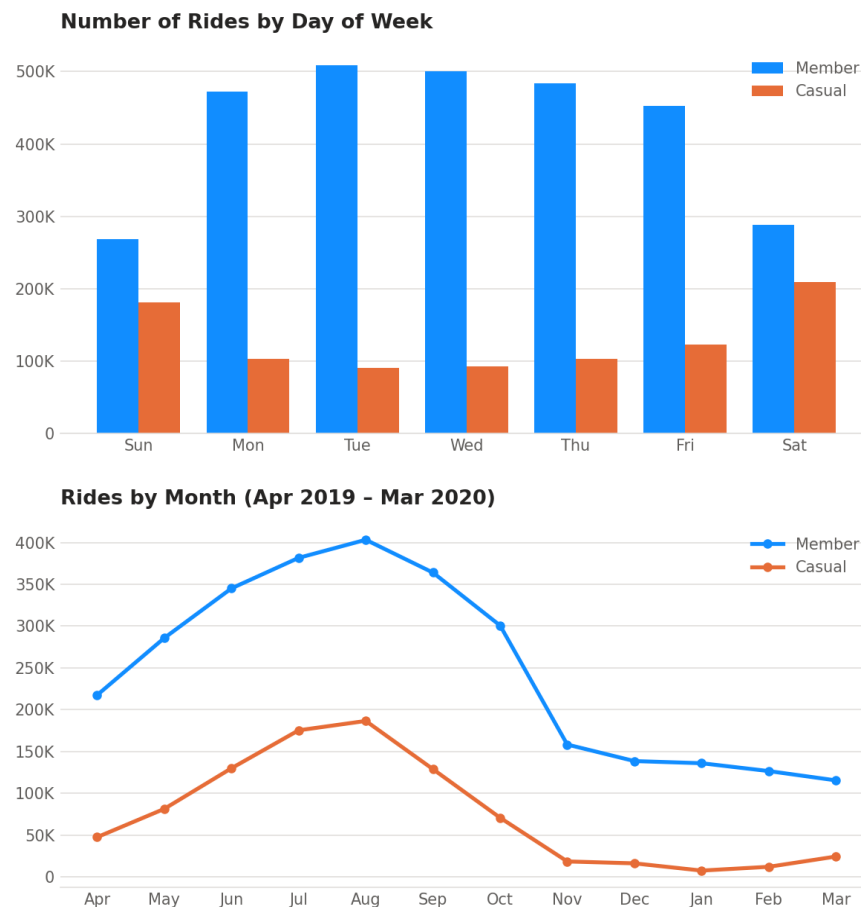
Cleaning. Removed maintenance/test trips and impossible durations (rides ≤ 0 or > 24h); 99.9% of rides retained.

What the data showed

	Member	Casual
Share of rides	76.8%	23.2%
Avg ride length	12.9 min	39.6 min

	Member	Casual
Median ride length	9.8 min	25.7 min
Busiest day	Tuesday	Saturday

Members ride short and often on weekdays — a commuting pattern. Casual riders ride about three times longer and peak on weekends — a leisure pattern. The two charts below show the split clearly.



Casual rides also concentrate at lakefront and tourist stations (Streeter Dr & Grand Ave, Lake Shore Dr & Monroe St, Shedd Aquarium, Millennium Park), versus downtown/transit-hub stations for members.

Recommendations

- **Weekend-framed membership offer, launched in spring** — sell the recreational, long-ride use casual riders already show, and time the push for the spring/summer surge when casual volume is highest.
- **Target the lakefront/tourist stations** where casual demand concentrates, rather than spending evenly across the network.
- **Sell the long-ride economics** — show casual riders at point of purchase what a membership saves on the ~40-min trips they already take.

Scope: full 12-month window (Apr 2019–Mar 2020). Data is public Divvy trip data, so no anonymisation was required.